BridgeCast AI: Breaking the Communication Barrier

Real-time Bidirectional Bridge Between Sign Language and Speech

13
Azure Services Deeply Integrated

300+
Sign Languages Supported

<800ms
Bidirectional Latency

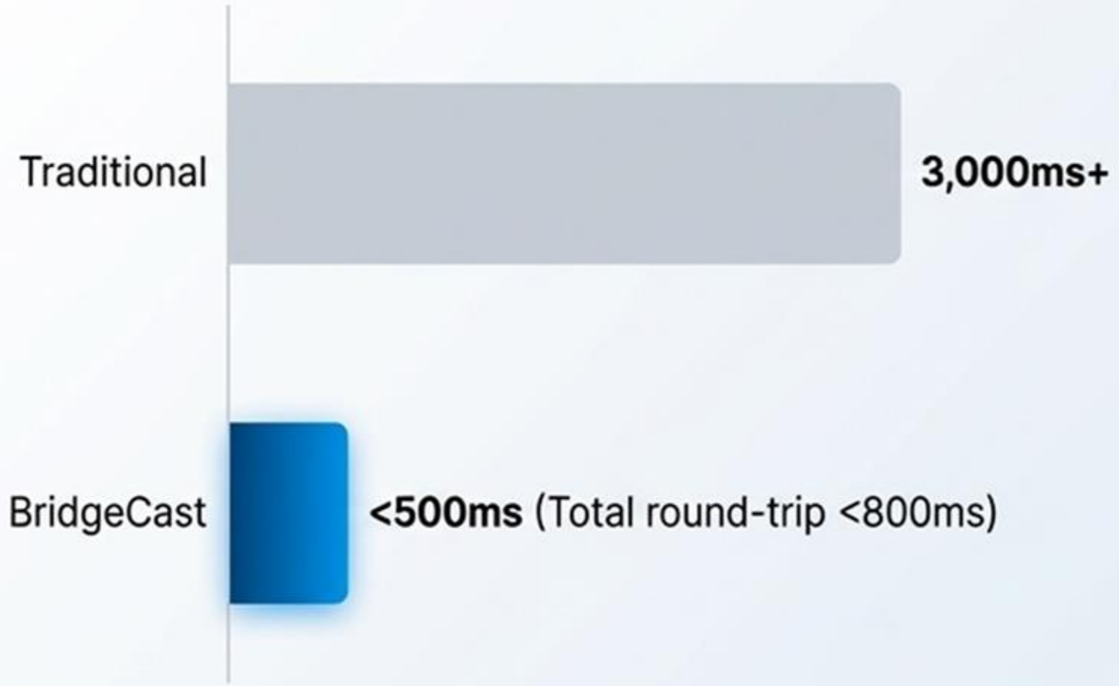
The Paradigm Shift: From Observer to Equal Participant

	Current State	BridgeCast AI
Role	Silent observer reading delayed captions.	**Full, equal participant from day one.**
Flow	One-way text-only.	**True bidirectional flow (Sign recognized, TTS plays voice).**
Nuance	Emotional context lost.	**Emotions preserved in both directions.**
Attribution	Missing from meeting notes.	**Speaker-attributed notes (`"Somi (sign): ..."`).**

Performance Metrics: Proven in Real-World Testing

Validated via 30-min live demo | 5 participants | Multilingual tech discussion

Latency Breakthrough



Accuracy & Reliability

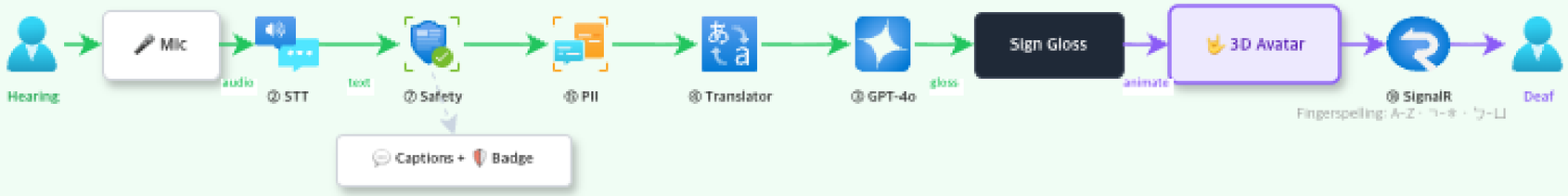
Achieved via Uni-Sign (ICLR 2025) model.
Running on Azure GPU VM (NC4as_T4_v3).
Pipeline: RTMPose → ST-GCN → mT5.

BridgeCast AI

Inclusive Communication Hub — Real-time Bidirectional Sign ↔ Speech · 15 Azure Services

ASL (139 signs + A-Z) · KSL (107 glosses + ㄱ-ㅎ) · TSL (20 signs + ㄅ-ㄣ) · Microsoft Innovation Challenge 2026

🗣️ Voice → 🤟 Sign (Hearing → Deaf)



🤟 Sign → 🗣️ Voice (Deaf → Hearing)

☁️ SHARED AZURE PLATFORM

🛡️ Responsible AI
- Content Safety · PII Redaction
- Fitzpatrick I-VI · WCAG 2.1 AA

430M+

people with hearing loss (WHO)

Tech Stack
FastAPI · PyTorch · MediaPipe
Three.js · Azure SDK · Bicep

15 Azure Services: ① App Service · ② Speech · ③ OpenAI · ④ Translator · ⑤ AI Vision · ⑥ GPU VM · ⑦ Content Safety · ⑧ Blob · ⑨ Comm Svc · ⑩ Functions · ⑪ Key Vault · ⑫ Monitor · ⑬ Cosmos DB · ⑭ SignalR · ⑮ AI Language

Azure Orchestration & Backend Scale

15 Azure Services Orchestrated

18 Custom Backend Services (FastAPI)

Data/State

- **Cosmos DB** — JetBrains Mono
- **Blob Storage** — JetBrains Mono

Real-time

- **Azure Speech** — JetBrains Mono
- **Communication Services** — JetBrains Mono
- **SignalR** — JetBrains Mono

AI/Logic

- **Azure OpenAI** — JetBrains Mono
- **AI Vision** — JetBrains Mono
- **AI Language** — JetBrains Mono

Security/Deploy

- **Key Vault** — JetBrains Mono
- **Azure Bicep (Infrastructure as Code)** — JetBrains Mono

AI Models & Real-Time Performance Validation

1	Model Engine	Uni-Sign **(ICLR 2025)**
2	Hardware Infrastructure	Azure GPU VM (NC4as_T4_v3)
3	Supported Vocabulary	**139** ASL signs | **107** KSL glosses (with dynamic fingerspelling fallback)

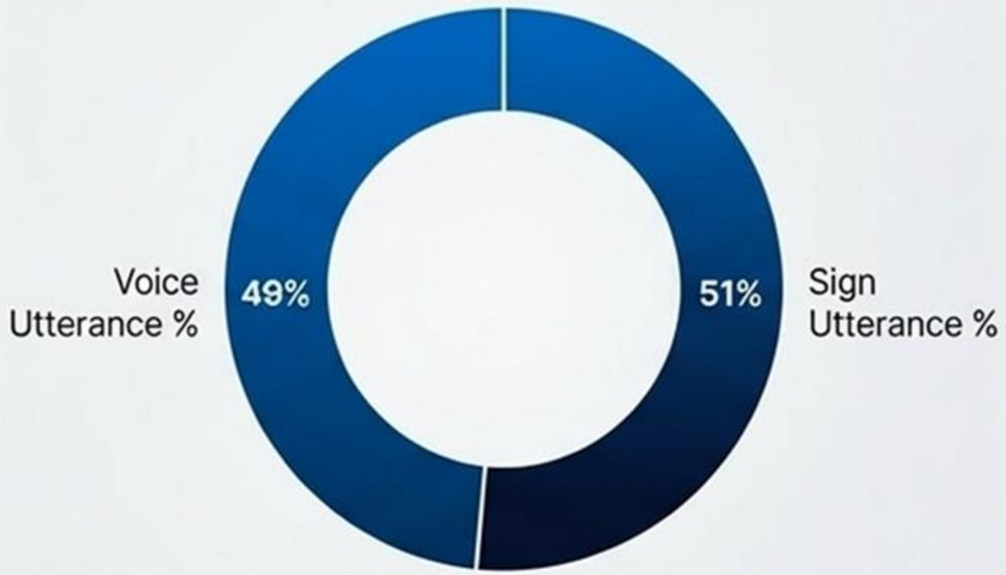
<500ms
Latency

87.3%
Sign Recognition Accuracy

95.2%
Speech Transcription Accuracy

Measurable Accessibility Impact

Participation Balance



Inter: Proving active participation over observation.

Bidirectional Success Rate

Tracking successful modality conversions.

Speaker Attribution

JetBrains Mono

```
Participant (sign): [Action Item Assumed]
```

Responsible AI & Global Scalability

Fairness
Modular support; tested across diverse skin tones.

Reliability
Graceful fallbacks and active confidence thresholds.

Privacy
PII auto-redacted before storage via AI Language.

Inclusiveness
WCAG 2.1 AA compliant.

Transparency
User-facing confidence scores.

Eccessibility
User-facing confidence scores.

Conatity
Auto-generated accessibility metrics.

Accountability
Auto-generated accessibility metrics.

Scalability Roadmap

Workplace → **Education** → **Healthcare** → **Public Services** →

NotebookLM

Team BridgeCast: Building the Equal Bridge

Let Every Voice Be Heard

Somi

Ollie Ni

TSAIHSUAN

- ✔ Acknowledging the Microsoft Innovation Challenge team
- ✔ Azure AI (for GPU VM support)
- ✔ Uni-Sign research community (ICLR 2025)
- ✔ Open-source contributors

Call to Action

GitHub: https://github.com/BridgeCastAI/BridgeCastAI

Demo: bridgecast.ai/demo

Docs: Comprehensive README, Responsible AI, and Transparency Note available in repo.